Project: Healthcare – Persistency of a Drug (Data Science Specialization)

Week 7: Deliverables

Team Member's Details

Group Name: DataVision
Name: Magdalena Ivanova
Email: magdalena.ivanova.ds@gmail.com
Country: Austria
College/Company: IMC Krems
Specialization: Data Science Specialization
Batch Code: LISUM49
Date: 18 October 2025

Table of Contents

---

1. Project Plan

Weeks	Date	Progress
Week 7	October 19, 2025	Project Initialization and Data Intake (Problem Description, Business Understanding, Data Intake Report)
Week 8	October 26, 2025	Data Understanding and Analysis of Data Issues (NA values, outliers, skewness)
Week 9	November 2, 2025	Data Cleansing and Transformation (apply ≥ 2 cleaning techniques, handle NA/outliers)

Week 10	November 9, 2025	Exploratory Data Analysis (EDA) and Recommendations
Week 11	November 16, 2025	EDA Presentation and Proposed Modeling Techniques
Week 12	November 23, 2025	Model Selection, Building, and Dashboard Creation
Final Week	November 30, 2025	Final Project Report, PowerPoint Presentation, and GitHub Submission

---

2. Problem Description

The pharmaceutical industry faces numerous challenges in ensuring patients remain adherent to prescribed medications. Patient non-adherence can negatively impact treatment outcomes, increase healthcare costs, and lead to further medical complications.

To address this, ABC Pharma Company has adopted a data-driven approach to identify factors influencing drug persistency. The goal of this project is to analyse patient-level data—including demographics, clinical characteristics, disease/treatment variables, and adherence metrics—to develop a predictive model that classifies patients as persistent or non-persistent with their prescribed drug.

The target variable ("Persistency_Flag") takes the value:

- 1 if the patient is persistent
- 0 if the patient is non-persistent

The project aims to provide actionable insights to healthcare providers and improve patient retention through data analysis and predictive modelling.

---

3. Business Understanding

Understanding why patients stop taking prescribed medication is crucial for both medical and business perspectives. Non-persistency not only leads to poor health outcomes but also affects the financial performance of pharmaceutical firms and insurance systems.

This project supports business decisions by:

- Identifying key variables affecting patient adherence.
- Predicting which patients are likely to discontinue treatment early.

- Enabling targeted interventions to improve persistency rates.
- Reducing costs related to non-adherence and treatment inefficiencies.

Ultimately, this data science solution can help improve patient outcomes, treatment effectiveness, and operational efficiency for healthcare organizations.

---

4. Project Lifecycle with Deadlines

1. **Week 7 (Oct 19, 2025)** – Submit PDF report containing problem description, business understanding, project lifecycle, data intake report, and GitHub link.
2. **Week 8 (Oct 26, 2025)** – Submit data understanding report: describe dataset, identify missing values, outliers, skewness, and propose handling approaches.
3. **Week 9 (Nov 2, 2025)** – Perform data cleansing and transformation; apply at least two cleaning/imputation methods and document them.
4. **Week 10 (Nov 9, 2025)** – Conduct exploratory data analysis (EDA), visualize insights, and provide business recommendations.
5. **Week 11 (Nov 16, 2025)** – Prepare EDA presentation for business users and include proposed modelling techniques for technical review.
6. **Week 12 (Nov 23, 2025)** – Build, evaluate, and compare models (linear, ensemble, boosting); create dashboard (if required).
7. **Final Week (Nov 30, 2025)** – Compile all work into the final report, PowerPoint presentation, and GitHub repository submission.

---

5. Data Intake Report

Name: Healthcare – Persistency of a Drug
Report Date: 18-10-2025
Internship Batch: LISUM49
Version: 1.0
Data Intake by: Magdalena Ivanova
Data Intake Reviewer: -
Data Storage Location: https://github.com/magdalena-ivanova-ds/Data-Glacier-Virtual-Internship/tree/main/week7

Dataset: Healthcare_dataset.xlsx

Total number of observations	3424
Total number of files	2 (Feature Description and Dataset)

Total number of features	27
Base format	.xlsx
Size of the data	899 KB

Proposed Approach:
There were no missing attributes or values in any of the datasets. The dataset will be analysed to identify relevant features affecting drug persistency, perform exploratory analysis, and apply classification models to predict patient adherence.

---

6. GitHub Repository Link

- https://github.com/magdalena-ivanova-ds/Data-Glacier-Virtual-Internship/tree/main/week7